



top Command in Linux

Introduction

Imagine you are using a **Linux system** and need to:

- Monitor CPU and memory usage in real time.
- Find out which process is consuming the most resources.
- Manage system performance efficiently.

How do you get a live view of system processes and resource usage?

This is where the **top (table of processes) command** helps!

Section 1: Learn

What is **top**?

The **top** command provides a real-time view of system processes, displaying CPU and memory usage for each process. It allows users to:

1. Monitor system performance continuously.
2. Identify high CPU and memory-consuming processes.
3. Manage and control running processes directly.

Why Do We Need **top**?

Without top	With top
Cannot check CPU or memory usage in real time	Provides live system resource monitoring
No way to find resource-hungry processes	Shows top CPU/memory consuming tasks



Without top	With top
Hard to troubleshoot system slowdowns	Identifies performance bottlenecks
No control over running processes	Allows process termination and priority adjustment

How Does **top** Work?

1. You run the **top** command in the terminal.
2. It continuously updates the display with live system process details.
3. You can interactively manage processes (kill, renice, filter).

An Interesting Anecdote

Before graphical system monitors (like Task Manager in Windows), Linux users relied on **top** for **real-time performance monitoring**. Even today, professionals prefer **top** because it is **lightweight and powerful**!

Section 2: Practice

Step-by-Step Guide to Using **top**

1. Running the **top** Command

```
# Start real-time process monitoring  
top
```

What happens here?

- Displays **live CPU, memory, and process usage**.
 - The list updates **every few seconds**.
-



2. Understanding the **top** Output

Field	Meaning
PID	Process ID
USER	Owner of the process
%CPU	CPU usage percentage
%MEM	Memory usage percentage
TIME+	Total CPU time used
COMM AND	Name of the running process

Why is this useful?

- Identifies **which processes are consuming system resources**.
 - Helps **troubleshoot system slowdowns**.
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3. Sorting Processes by CPU Usage (**P**)

While top is running, press:

P # Sort by CPU usage

How does this help?

- Lists **highest CPU-consuming processes first**.
-

4. Sorting Processes by Memory Usage (**M**)

While top is running, press:

M # Sort by Memory usage



What does this do?

- Displays **which applications use the most RAM**.
-

5. Killing a Process from **top (k)**

While top is running, press:

k # Enter PID and confirm termination

Why is this important?

- Stops **unresponsive or resource-hogging applications**.
 - Helps **free up system resources**.
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6. Changing Process Priority (**r**)

While top is running, press:

r # Enter PID and new priority (-20 for highest, 19 for lowest)

Why is this useful?

- Adjusts **priority of tasks dynamically**.
 - Ensures **critical applications get more CPU power**.
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7. Filtering Processes (**o**)

While top is running, press:

o # Enter filter criteria (e.g., USER=root to show root processes only)

How does this help?

- Focuses **only on relevant processes**.
- Useful for **multi-user systems**.



8. Exiting **top**

```
# Press:
```

```
q # Quit top
```

What does this do?

- Closes the **top** command and returns to the terminal.

Real-World Example

A **system administrator** notices **slow server performance**. They need to:

1. Find which process is consuming too much CPU.
2. Terminate an unresponsive program.
3. Increase priority for an important background task.

What did they do?

1. Used **top** and pressed **P** to identify high CPU-consuming processes.
2. Pressed **k**, entered the PID, and killed an unresponsive process.
3. Pressed **r**, entered the PID, and increased priority for a critical task.

Result? The **server performance** improved immediately!

Section 3: Know More

Frequently Asked Questions

1. How do I run **top** for a specific user?

```
top -u username
```

2. What is the difference between **top** and **htop**?



- **top** is default and lightweight.
- **htop** is interactive, colorful, and user-friendly.

3. How do I display process tree in **top**?

```
V # Press 'V' while top is running
```

4. How do I refresh **top** faster or slower?

```
d # Press 'd' and enter new refresh time in seconds
```

5. Where can I practice **top** commands?

- Use a **Linux Virtual Machine**.
- Try **server monitoring on AWS EC2**.

Conclusion

The **top** command is an essential tool for monitoring and managing Linux system performance in real-time.

Start by **checking system resource usage**, **sorting processes**, and **adjusting priorities**, and soon you'll be **optimizing Linux like an expert!**